जम्मू केंद्रीय विश्वविद्यालय

Central University of Jammu

रह्या-सुचानी (बगला) , जिला सांबा- 181143, जम्मू (जम्मू और कश्मीर)

Rahya-Suchani (Bagla), District: Samba -181143, Jammu (J & K)

07 march 2025

Roll No.												

MID SEMESTER EXAMINATIONS

Programme	:	**B.Tech. (CSE)**	Semester	:	**6th**
Course Name	:	**Data mining and Warehousing**	Course Code	:	**BECSE0O011**
Duration	:	**2 Hours**	Credits	:	**04**
Maximum Marks	:	**60**	Minimum Marks	:	

After pursuing this course, the student will be able to:	
CO01	Understand data warehousing concepts, architecture, and implementation.
CO02	Design and manage data warehouses to support efficient data analysis and retrieval.
CO03	Apply data mining techniques to discover patterns, clusters, and associations.
CO04	Evaluate and improve data mining models for better accuracy and knowledge extraction.
CO05	Solve real-world business problems using data mining and warehousing methods.

Part A Very Short Answer Type Logical Questions of 04 marks related to different COs		Total Marks 4 X 4 = 16	
S. No.	**Statement**	**Level**	**CO**
A1	Discuss the various characteristics of Datawarehouse.	1	1
A2	What are the various tasks performed in data pre-processing?	1	3
A3	Design a star schema for a healthcare data warehouse comprises of four dimension tables.	3	1,2
A4	How does ETL contribute to data warehouse development?	2	1
Part B Short Answer Type Analytical Questions of 06 marks related to different COs (Attempt any Four questions)		Total Marks 4 X 6 = 24	
B1	What is data cube.? Create and explain a 3-D data cube representation of the sale data according to time, item, and location dimensions.	3	2
B2	What are the various kind of data that can be mined and what kind of patterns can be mine from such data?	1	2
B3	Draw and explain Star Schema and Snowflake Schema with suitable example.	3	1,2
B4	Discuss various techniques for dealing with missing data in database.	1	3
B5	What are various approaches for building data warehouse and also discuss the steps involve in data warehouse design process.	2	2
Part C Long Answer Type Descriptive Questions of 10 marks related to different COs (Attempt any Two questions)		Total Marks 2 X 10 = 20	
C1	a) Differentiate between OLTP (Online Transaction Processing) and OLAP (Online Analytical Processing) systems. b) Explain the architecture of data mining process.	2	1,3
C2	Explain various OLAP operations with suitable examples.	3	1,2,3
C3	a) Draw and explain three tier architecture of Data warehouse. b) What is KDD? Explain various steps involve in the KDD process.	2	1,3